

The Attralucian Essays:

Exploring the Finite



First Edition

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The Attralucian Essays



Before Artificial General Intelligence:
Language, Measurement and Intelligence in
the Age of AI

Kevin R. Haylett

Overview

Artificial General Intelligence has become one of the defining ambitions of the present technological age. Yet despite extraordinary advances in machine learning and language modelling, remarkably little agreement exists concerning the thing that researchers ultimately hope to construct. The word *intelligence* appears familiar, but its historical uses reveal considerable semantic uncertainty. It has referred to intellect, reason, judgement, wisdom, adaptation, learning, problem-solving, measurable aptitude, computational performance and, increasingly, economic productivity.

This essay argues that intelligence is not encountered directly as a substance residing within a person, animal or machine. It is registered through finite interactions and subsequently attributed to a system by observers working within inherited languages, practices and systems of measurement. The modern intelligence score compressed a complex interaction into a number. The Turing test replaced an uncertain question about machine thought with a finite linguistic procedure. Contemporary definitions of Artificial General Intelligence similarly translate intelligence into task performance, generality, autonomy and economic value.

These operations are useful, but they do not resolve the prior question of what intelligence is. They reveal in-

stead that the category of intelligence is constructed, measured and communicated through language. Drawing upon Wittgenstein’s account of language as part of a “form of life”, the essay proposes that a form of life may be understood as a dynamical system composed of organisms, environments, practices, instruments, symbols, memories and institutions. Language emerges within this system and then acts back upon it, stabilising some distinctions while restricting or redirecting others.

Within a Geofinite framework, meaning is approached as a historically constrained family of functional symbolic trajectories. Intelligence is correspondingly treated not as an unqualified possession but as a judgement formed from measurable interactions between a system, its environment, its history and the symbolic framework through which its conduct becomes distinguishable. A coherent account of Artificial General Intelligence may therefore require something presently missing: a developed model of language, registration, semantic uncertainty and symbolic change.

The Question Before Artificial General Intelligence

Every scientific period inherits words whose familiarity conceals unresolved problems. Some of these words become so deeply established that they appear to name plainly observable things. They pass between ordinary speech, scientific writing, education, government and commerce without their meaning being reconsidered at every use. *Intelligence* is one such word.

We describe a child as intelligent, an animal as intelligent, a decision as intelligent, a military service as intelligence, and a computer system as artificially intelligent. We speak of emotional intelligence, collective intelligence, practical intelligence, general intelligence and superintelligence. The word may indicate quickness, understanding, adaptation, foresight, memory, verbal facility, originality, judgement or successful action. These uses overlap, but they are not identical.

The apparent unity of the term can therefore be misleading. It may suggest that all these uses refer to a single underlying quantity or faculty when they may instead arise from a family of related judgements made under different conditions.

This uncertainty has become especially important because Artificial General Intelligence, usually abbreviated

to AGI, is no longer merely a speculative subject. It is invoked in research programmes, corporate missions, policy discussions, investment decisions and predictions about the future organisation of society. The ambition to create AGI is presented as an engineering goal, even though the property to be engineered remains unsettled.

Modern artificial intelligence systems have unquestionably achieved remarkable capabilities. They recognise images, transcribe speech, translate languages, generate software, predict molecular structures, control machinery and participate in extended written conversations. Large language models can produce responses of a kind that, within a previous technological period, would readily have been treated as evidence of substantial human intellectual ability.

These achievements create a powerful temptation. As machine performance expands across tasks, it seems natural to imagine that the system is moving along a scale towards general intelligence. Sufficient performance, sufficiently widely distributed, will eventually cross a threshold. On one side there is specialised artificial intelligence; on the other there is AGI. Yet this apparently straightforward picture depends upon several unresolved assumptions.

What counts as a task? Who determines whether two tasks belong to the same domain? What makes a capability general rather than specialised? What form of

continuity must exist between one performance and another? Does a system possess the capability between demonstrations, or is the capability registered only when a suitable interaction occurs? Whose judgement establishes successful performance? Which human abilities are being used as the reference? And what is preserved when all these judgements are compressed into a benchmark score?

Before asking whether a machine possesses general intelligence, we must therefore ask how the human category of intelligence was formed, stabilised and measured. That question cannot be answered by looking only at present machines. It requires a historical account of the word, the practices surrounding it, and the changing systems in which intelligent behaviour has been distinguished.

The following history is necessarily selective. It concentrates on the European and Anglo-American symbolic trajectory from which contemporary computer science and AGI discourse largely emerged. It is not offered as a universal history of how every culture has understood intelligent conduct. Its purpose is narrower: to identify the inherited language now being used to define and construct artificial intelligence.

Intelligence Before Measurement

Long before intelligence became a psychological score, human communities distinguished between different forms of successful perception, memory, judgement and action. They recognised the skilful hunter, persuasive speaker, wise elder, capable craftsman, reliable navigator and thoughtful ruler. These capacities were not necessarily treated as manifestations of one measurable property. They were embedded in particular practices.

Ancient Greek thought contained several partially overlapping terms for mental and practical capacities. *Nous* could refer to intellect or apprehension; *logos* could indicate speech, account, relation or reason; *phronesis* concerned practical judgement; *sophia* was associated with wisdom; and *techne* with skilled making. These words did not form a modern psychological classification. They belonged to a wider system of distinctions concerning conduct, knowledge, order, speech and human excellence. The history of *nous* alone is complex, and its significance cannot be reduced to the modern English word *intelligence*.

This older plurality is significant. A person might be capable in one activity and deficient in another without the difference being expressed as variation in a single general quantity. Judgement was often inseparable from the situation in which it was demonstrated.

A wise decision was wise in relation to circumstances. A skilful action was skilful within a practice. A convincing argument depended upon an audience and a language. The action, setting and judgement remained comparatively close together.

As classical and later European traditions developed, distinctions between intellect, reason, understanding, memory, will and imagination acquired different places within religious, philosophical and educational systems. These terms were debated, translated and reorganised over centuries. Their meaning changed as they moved between Greek, Latin and later vernacular languages.

The Latin *intelligentia* carried associations with understanding or discernment. It did not originally name a standardised quantity that could be assigned to every individual. Intelligence was closer to an activity or capacity of understanding than to a score.

During the early modern period, transformations in mechanics, astronomy, mathematics and instrumentation altered the language through which the world was described. The seventeenth century was especially important. A comparatively small and interconnected group of natural philosophers established symbolic and mathematical trajectories that subsequently became foundational to modern science.

The natural world was increasingly represented through

measurement, geometry, mechanism and mathematical relation. The success of these methods encouraged an expanding confidence that apparently complex phenomena could be decomposed into simpler operations and represented through formal structures.

Human thought itself became open to mechanical comparison. Hobbes described reasoning in terms resembling calculation. Descartes compared animals and bodies with machines while reserving a special role for human thought. Leibniz imagined formal systems in which disputes might be resolved through calculation. None of these positions was identical to modern artificial intelligence, but together they contributed to a growing expectation that mental operations might be describable as ordered procedures.

This was not merely the discovery of a previously hidden mechanical mind. It was also the formation of a new symbolic language through which thought could be re-described. When a clock is used as a model for an organism, or calculation as a model for reasoning, the comparison may illuminate important regularities. But it also carries the structure of the instrument into the thing being described. The model selects some features and suppresses others. The early modern period therefore did more than advance knowledge of the world. It helped establish a long-running relationship between intelligence, formal procedure and mechanism.

Industrialisation strengthened this tendency. Expanding states, armies, schools, factories and administrative systems required people to be selected, ranked and assigned to roles. Differences between individuals became matters of institutional concern. It was no longer sufficient to observe that one person appeared thoughtful and another quick. Large organisations required portable judgements. The conditions were forming for intelligence to become measurable.

At the same time, evolutionary thought encouraged a different but related understanding. Intelligent behaviour could be considered in relation to adaptation: the ability of an organism to respond successfully to its surroundings. This widened the category beyond abstract reasoning. Animals, organisms and eventually machines could be compared through their capacity to learn, adjust and survive. These historical trajectories—reason, mechanism, administration, adaptation and measurement—did not merge into a single settled account. They remained partially distinct. Yet they provided the symbolic background from which the modern scientific object called *intelligence* could emerge.

The Century of Measurement

The late nineteenth and early twentieth centuries transformed intelligence from a broad human judgement into

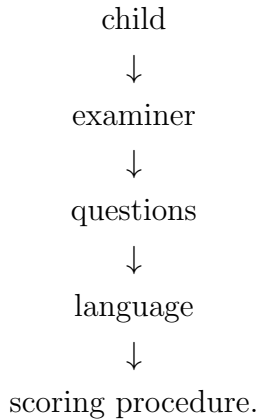
an object of systematic measurement. This development was part of a wider movement. Human characteristics were increasingly counted, classified and statistically compared. Height, weight, reaction time, sensory discrimination, school performance and social behaviour could be entered into tables and examined for regularities.

The attraction of such methods is easy to understand. Informal human judgement is variable. It may be prejudiced, inconsistent or impossible to compare across institutions. A standard procedure promises fairness, reproducibility and administrative usefulness. Yet the creation of a standard measure also changes the object being measured.

Alfred Binet and Théodore Simon developed their early intelligence scale in France during the first years of the twentieth century. Its practical purpose was educational: to help identify children who might require additional assistance. The Binet–Simon scale became a foundation for later intelligence testing, including the Stanford–Binet revisions [1, 2].

The procedure involved a structured interaction between a child and an examiner. The child listened, interpreted instructions, remembered material, manipulated objects, answered questions and responded to a particular adult in a particular setting. The examiner judged the responses according to defined criteria. The resulting assessment did not arise from the child alone. It arose from

a coupled arrangement:



The score was subsequently attributed to the child. This compression was institutionally useful. A many-layered interaction became a value that could be recorded, compared and transmitted. But the usefulness of the value can obscure the trajectory from which it arose. The measurement depended upon language comprehension, familiarity with the setting, attention, confidence, cultural experience, the examiner’s conduct and the design of the tasks. These were not incidental additions to an otherwise pure intelligence. They were conditions of the measurement.

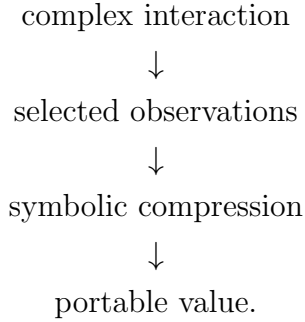
Charles Spearman’s influential 1904 paper sought to determine and measure “general intelligence” objectively. He examined correlations between different forms of performance and proposed that a common factor contributed across them. Spearman himself distinguished between the reliability of a measurement and the separate ques-

tion of whether the measured form deserved the name *intelligence*. That caution is important: the existence of a stable statistical relation does not by itself settle the meaning of the word assigned to it [3].

Over time, however, the distinction between measurement and object often became compressed. A test designed to register performance under specified conditions could be treated as though it disclosed an internally stored quantity.

The development of intelligence testing brought genuine practical achievements. Tests could help identify educational needs, examine patterns of cognitive difficulty and support certain forms of comparison. Yet their expansion into schools, military selection, employment and social policy also gave the score an authority extending far beyond the original interaction. A number can travel more easily than the conditions that produced it. It can enter a file, a database, a statistical study or an administrative decision. It may acquire an appearance of independence because its provenance is no longer visible at the moment of use. The intelligence score therefore exemplifies a broader feature of symbolic measurement:

Exploring AGI



The compression is not necessarily illegitimate. Every practical measurement requires selection. No measurement carries the total history of the measured interaction in its immediately visible result. The problem arises when the compressed symbol is treated as complete. A Geofinite account insists that a measured value retains, whether explicitly shown or not, the history of its production. The score is not a free-standing essence. It is a finite registration created through a procedure.

This distinction becomes crucial when psychological measurement is later used as a template for machine intelligence. If human intelligence has already been converted into performance across selected tasks, then a machine that performs well on similar tasks may appear to possess the same underlying property.

However, it may instead have entered the same measurement arrangement. The machine is not necessarily revealing a hidden human-like quantity. It is producing outputs that satisfy inherited criteria. This is not a dis-

missal of machine capability. It is a clarification of what has been measured.

Turing’s Elegant Substitution

Alan Turing’s 1950 paper *Computing Machinery and Intelligence* marks one of the most important transitions in the history of machine intelligence.

Turing begins with the question, “Can machines think?” He immediately recognises that defining *machine* and *think* by ordinary usage would create serious difficulty. Rather than attempting to settle both terms, he replaces the original question with a more precisely arranged interaction: the imitation game [4].

This move was elegant because it avoided dependence upon an inaccessible inner condition. Turing did not require an observer to open the machine and identify a substance called thought. Instead, he proposed a finite communicative procedure involving a machine, human participants, questions, responses and judgement.

The question changed from:

Can a machine think?

to something closer to:

Under specified linguistic conditions, can an evaluator reliably distinguish the machine’s

responses from those of a human participant?

This is not a trivial alteration as it changes the object of measurement. The imitation game measures a result produced across an interaction. It does not directly measure an independently established quantity residing inside the machine. In this respect, Turing's procedure is compatible with an important Geofinite principle: when a term cannot be measured directly, construct a finite interaction and state what is actually registered.

Turing's method is often interpreted as though he had proved that successful linguistic imitation and intelligence were identical. That reading is too simple. His paper is exploratory and addresses numerous objections. The imitation game functions as a replacement for an unstable question, not as the discovery of a final substance.

Nevertheless, later culture transformed the imitation game into the "Turing test", and the test acquired a more definite symbolic role. Passing it came to be treated as a possible threshold between machine performance and machine intelligence. Again, a trajectory was compressed. The original paper's careful reframing became a cultural marker.

This history reveals something fundamental. The modern discussion of machine intelligence was linguistic from the beginning. The machine's capacities became accessible to judgement through conversation. The evaluator

asked questions. The machine generated symbols. The evaluator interpreted those symbols through an inherited human language. The result depended upon the evaluator's expectations concerning how a human would respond. The procedure therefore contained at least three interacting models:

1. the machine's model of linguistic continuation;
2. the human participant's use of language;
3. the evaluator's model of human and machine behaviour.

The test did not stand outside language and then measure intelligence neutrally. Language was the medium of the test, the material of the responses and part of the evaluator's criterion.

This is precisely why Turing remains so important. His proposal did not merely anticipate conversational machines. It exposed, perhaps unintentionally, the dependence of machine intelligence upon a linguistic measurement system.

Artificial Intelligence and the Modern Epoch

The term *artificial intelligence* was established through the proposal for the Dartmouth Summer Research Project,

written in 1955 by John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon. The proposal was based upon a bold conjecture: that every aspect of learning or intelligence could, in principle, be described precisely enough for a machine to simulate it. The proposed topics included language, abstraction, concept formation, problem-solving and self-improvement [5]. This conjecture contains the foundational confidence of the field.

First, intelligence is assumed to be decomposable into aspects. Second, those aspects are assumed to be describable. Third, sufficiently precise description is assumed to permit machine simulation. Each step depends upon language.

To divide intelligence into aspects requires names and boundaries. To describe an aspect precisely requires a symbolic system. To determine whether a machine simulates it requires criteria of correspondence. The technical programme therefore began inside a prior linguistic programme, even where that prior dependence was not made explicit.

Early artificial intelligence research pursued several approaches. Some researchers attempted to represent reasoning through symbols and rules. Programs were developed for theorem proving, game playing, planning and language use. Later expert systems encoded specialist knowledge in the form of facts and conditional rules. These systems could perform impressively within restricted

domains but often became fragile when moved beyond their carefully constructed environments. The history of the field subsequently included cycles of optimism, disappointment and reduced funding commonly called AI winters [6].

Another trajectory developed through artificial neural networks and machine learning. Rather than specifying every relevant rule in advance, researchers constructed systems whose internal parameters changed in response to data.

The distinction between symbolic and learning approaches was never absolute, but the emphasis differed. One sought to write intelligence into formal procedures; the other sought to generate useful behaviour through training. Both remained dependent upon prior human selections. Someone chose the architecture. Someone assembled or permitted the training data. Someone specified an objective. Someone determined the acceptable output. Someone selected the test.

The system's behaviour could be emergent relative to its designers' immediate expectations without being independent of the larger symbolic and material arrangement that produced it. The introduction of the Transformer architecture in 2017 was a major technical development. Designed initially for sequence processing and machine translation, it used attention mechanisms to model relations within sequences more efficiently than many previ-

ous architectures [7].

Transformer-based language models could then be trained on extremely large collections of text. By 2020, GPT-3 demonstrated that increasing model scale and training across large textual corpora could produce strong performance across a wide range of tasks through prompting and a small number of examples. Tasks could be presented to the same model through language rather than requiring a separately engineered system for each one [8]. This development changed the practical appearance of generality.

A single linguistic interface could now support translation, summarisation, question answering, classification, code generation, explanation, imitation and creative writing. The system did not merely produce one kind of output. It could enter many human language-games.

The appearance of breadth was intensified because ordinary language acts as a universal routing system for human activities. A user can describe a legal problem, mathematical exercise, programming task, personal dilemma or historical question in the same broad medium.

Consequently, a system able to continue language convincingly may appear able to move directly between the domains represented by language.

This is a profound capability. But it also creates a measurement ambiguity.

Has the system acquired general intelligence?

Or has it acquired broad access to the symbolic interface through which humans describe and coordinate many specialised activities?

These possibilities are not mutually exclusive. The second may contribute substantially to the first. The important point is that they should not be treated as identical without examination.

AGI as an Epoch-Specific Construction

Contemporary definitions of AGI reveal the practical concerns of the institutions that produce them. OpenAI’s published Charter defines AGI as highly autonomous systems that outperform humans at most economically valuable work. Google DeepMind’s proposed “Levels of AGI” framework classifies systems through breadth or generality, depth of performance and questions of autonomy [9, 10].

These definitions are operationally valuable. They can guide research, benchmarking, risk assessment and institutional decisions. Consider:

“Economically valuable work” places intelligence within the organisation of labour, markets and production.

“Performance” places intelligence within defined tasks and scoring systems.

“Generality” concerns the breadth of successful performance across categories selected by human communities.

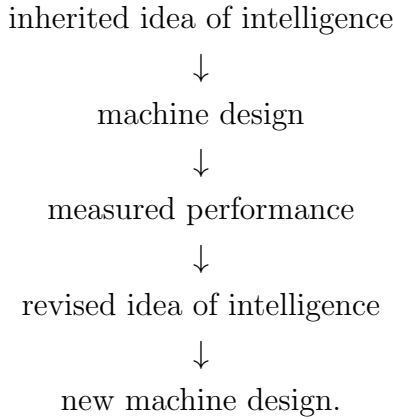
“Autonomy” concerns how far a system acts without immediate human intervention, but does not by itself establish the source or character of the system’s intelligence.

A system might be highly autonomous but inflexible. Another might be deeply dependent upon human interaction yet participate in an exceptionally productive coupled intelligence. The present AGI language therefore reflects the current epoch: industrial, computational, economic and benchmark-driven. This does not invalidate it. Every definition arises from a form of life. The difficulty appears when a local definition is treated as a universal discovery.

A society organised around different priorities might recognise intelligence through ecological balance, continuity of memory, social coordination, restraint, practical care or survival under changing environmental conditions. A non-human species might divide relevant activity in ways inaccessible to human benchmarks. A machine with unfamiliar sensory relations might develop successful distinctions for which human language has no established category.

AGI is consequently not only a machine that researchers hope to build. Rather it is also a symbolic category that institutions are actively constructing. The category in-

fluences funding, architecture, benchmarking and public expectation. The resulting systems then alter the category that guided their construction. From this perspective, a recursive trajectory develops:



This is not a defect that can simply be removed. It is how symbolic and technical systems develop. But its history must remain visible if we are to understand what is being claimed.

The Forgotten Foundation: Language

Language is often treated as one capability among several that an intelligent agent may possess.

A list might include perception, memory, reasoning, planning, learning, movement and language. Under this arrangement, language is a module or channel through which

an already formed intelligence communicates. The development of large language models invites a different possibility. Language may not merely express a previously completed intelligence. It may be one of the principal systems through which human intelligence is constructed, coordinated, preserved and extended.

A human child does not first acquire a complete private model of the world and then attach words to it. The child develops through interaction with people, objects, gestures, sounds, corrections, repetitions and institutions. Linguistic categories help stabilise distinctions. They preserve prior registrations and guide future attention.

Language is therefore not simply a list of names. It is a historically accumulated system of executable distinctions. Words enable people to separate, connect, remember, instruct, prohibit, compare, imagine and revise. Sentences carry actions across time and between individuals. Written symbols permit trajectories to continue beyond the memory or lifespan of a single person. Mathematics, law, administration and science all depend upon such continuity. This does not mean that every form of intelligence is linguistic. Animals navigate, learn, cooperate and solve problems without human language. Human sensorimotor activity also exceeds what can be stated at any moment. The stronger and more defensible claim is this:

Complex, cumulative, publicly examinable in-

telligence depends upon systems of symbolic registration.

Without symbols, an interaction may still alter a system. But it cannot be preserved, compared and transmitted with the same degree of precision across people and generations. Language makes the history of a distinction available for further use.

Wittgenstein and the Limits of Language

Ludwig Wittgenstein's work provides a valuable historical bridge into this question.

In the *Tractatus Logico-Philosophicus*, Wittgenstein wrote:

“The limits of my language mean the limits of my world.”

The sentence should not be reduced to the crude claim that words manufacture every physical interaction. Its force concerns the world that can be expressed, distinguished and held within the speaker's language [11].

A person can encounter something for which no adequate word exists. The interaction does not therefore vanish. But its preservation and communication become difficult. It may be absorbed into an existing category, registered incompletely or forgotten.

The limits of language are therefore not necessarily the limits of all interaction. They are limits upon the cur-

rently accessible symbolic world.

Wittgenstein's later work moved away from the idea that language could be understood principally as a formal picture of facts. He drew attention instead to language-games, practices and use.

For a large class of cases, he suggested, the meaning of a word is its use in the language. He also wrote that to imagine a language is to imagine a form of life, and that speaking a language is part of an activity [12].

These remarks are valuable not because Wittgenstein must be treated as the final authority on language. His position within the philosophical canon is itself the result of a particular academic and historical trajectory. The remarks are valuable because they direct attention away from the idea that meaning resides as a fixed content inside an isolated word.

Words live within activity. They are taught, corrected, repeated and adapted. Their use depends upon shared practices and expectations.

The term *intelligence* does not carry the same function in a classroom, animal-behaviour laboratory, military organisation, computer-science paper or corporate strategy document. The visible word remains similar while the surrounding language-game changes. The later Wittgenstein therefore provides an opening. But a Geofinite framework can take the argument in a more explicitly

dynamical and measurable direction.

A Form of Life as a Dynamical System

Wittgenstein's phrase "form of life" has generated many interpretations, partly because he used it sparingly and did not supply a complete formal definition [13].

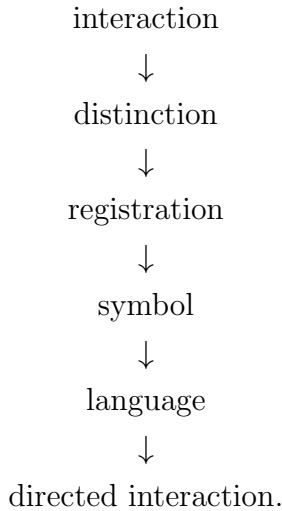
Within a Geofinite framework, a form of life can be approached as a dynamical system.

It is composed not only of speakers and words but of:

- organisms;
- bodies and sensory systems;
- physical surroundings;
- repeated practices;
- instruments;
- institutions;
- memories;
- written records;
- material resources;
- social expectations;
- technologies;

- and established symbolic structures.

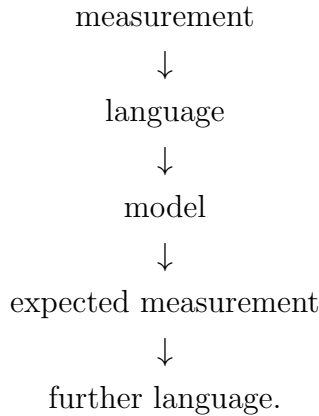
These components interact over time. Some interactions repeat and stabilise. Others weaken, bifurcate or disappear. New instruments create new distinctions. New distinctions require new symbols. New symbols alter the measurements that later become possible. Language is generated within this dynamical system, but it subsequently becomes an active part of the system. The relation is recursive:



The first stages need not always be consciously separated. An organism encounters a difference. The difference alters its behaviour. Repeated interactions stabilise a distinction. A gesture, sound or mark becomes associated with that distinction. The symbol can then be reused in the absence of the original interaction. Once established,

the symbol directs future attention.

A person taught the word *intelligence* begins to classify conduct through that word. A teacher notices an intelligent answer. A researcher designs an intelligence test. A company advertises intelligent software. The inherited symbol changes which events become salient. This produces another loop:



The loop is productive. Without it there would be no cumulative science, education or shared technical practice.

But it also introduces constraint. Once a symbolic basin becomes deep, new interactions are likely to be interpreted through it. The language prepares the user to find certain structures and overlook others.

Meaning as a Stabilised Trajectory

In a static account, a word is imagined to possess a meaning:

$$M(w) = m. \tag{1}$$

The word w is treated as though it contains or points reliably to a stable item m . A dynamical account requires a different form:

$$M(w, t, H, C), \tag{2}$$

where the registered meaning of the word depends upon time t , inherited history H , and the interactional context C . This does not imply that words have no stability. Communication would be impossible if every use were entirely novel. Stability exists through repeated trajectories.

A word is taught within examples. It appears in familiar grammatical arrangements. It is attached to practices. Departures are corrected. Documents preserve earlier uses. Institutions enforce some meanings more strongly than others. The word consequently develops a basin of likely continuations. Meaning can therefore be approached as a historically constrained family of functional symbolic trajectories.

A Functional Symbolic Trajectory is not merely a sequence of marks. It is a finite pathway through which symbols carry distinctions, constraints and expectations across an interaction. A trajectory may be spoken, written, mathematical, gestural or computational. It has a beginning, development and practical continuation.

The phrase *artificial general intelligence* is itself such a trajectory. It activates inherited associations:

- *artificial* suggests manufacture rather than biological development;
- *general* suggests breadth beyond a narrow task;
- *intelligence* carries the accumulated history of reason, testing, adaptation, performance and judgement.

The combined term appears precise because each word is familiar. Yet their combination inherits the semantic uncertainty of all three.

Semantic Thickness

In Geofinitism, no finite symbol carries perfectly complete and universally identical meaning.

A symbol has semantic thickness. This thickness is the range of partially overlapping trajectories in which the symbol may participate while still being recognised as the same symbol.

The thickness of *intelligence* is considerable.

It includes at least:

- understanding;
- rapid learning;
- memory;
- reasoning;
- linguistic fluency;
- adaptation;
- foresight;
- problem-solving;
- practical judgement;
- creativity;
- social coordination;
- test performance;
- and economic usefulness.

Different users activate different portions of this thickness. Disagreement about whether a language model is intelligent may therefore arise even when participants agree about the observed behaviour. They may be applying different inherited trajectories of the word.

One person means verbal and analytical competence. Another means embodied adaptation. Another requires con-

sciousness. Another requires independent goal formation. Another uses intelligence to mean measurable task performance.

The argument appears to concern the machine, but part of it concerns the unstated language of the observers. This is why definition alone cannot fully solve the problem. A new definition is itself another symbolic registration. It may be useful, but it does not erase the histories carried by existing use.

What is required is provenance: an account of how the term has been constructed, which measurements support it, what has been compressed, and under which conditions the judgement remains admissible.

Intelligence as Interaction

The ordinary grammar of intelligence encourages us to treat it as a possession. A person *has* intelligence. An animal *has* intelligence. A machine either *has* or *does not have* intelligence. This grammar directs attention inward. We look for the property inside the bounded object. Yet intelligence is not observed there. We observe conduct.

A person answers a question, solves a problem, learns a procedure, notices a pattern, repairs an error or changes strategy. An animal navigates an environment, uses an object, coordinates with others or adapts to altered con-

ditions. A machine classifies an image, continues a sentence, controls a process or produces a successful plan.

From these finite interactions, an observer registers intelligence. The judgement may be well supported. But it remains a judgement formed from interaction. A Geofinite reconstruction can therefore begin with the following proposition:

Intelligence is initially accessible as a class of measured interactions, not as an independently measured substance possessed by an isolated entity.

This does not deny that systems have internal structure. Internal memory, organisation and prior learning clearly affect future conduct. The claim is narrower: whatever internal structure exists becomes publicly accessible through finite interactions and measurements. An observer never measures “intelligence in itself”. The observer measures selected changes and outcomes.

A Conditional Registration

A provisional expression might be written as:

$$I_M(S, E, H, L, P), \tag{3}$$

where:

- I_M is a measured or registered judgement of intelligence;
- S is the system under consideration;
- E is the interactional environment;
- H is the accessible history of the system and measurement;
- L is the linguistic or symbolic framework;
- P is the procedure through which the judgement is produced.

There is no unqualified I detached from these conditions.

Change the environment and the measured performance may change. Change the language and the task may change. Change the procedure and the criterion may change. Remove the history and an adaptive response may appear inexplicable. This conditionality is not a weakness. It is an honest account of measurement. Consider a person unable to answer a question in an unfamiliar language. The failure does not establish the absence of all intelligence. It establishes a failure within a particular symbolic interaction.

Consider an animal unable to complete a human-designed puzzle while navigating its natural environment with extraordinary success. The test may register a mismatch between systems rather than a general deficiency. Also consider a language model that performs impressively in

text but cannot maintain stable action in a physical setting. The contrast reveals the boundaries of the measured system. It does not require either total celebration or total dismissal. Intelligence becomes a profile of trajectories rather than a single metaphysical rank.

Continuity, Revision and Reach

Within this interactional account, several features become especially important. An intelligent system may:

- retain the effects of prior interactions;
- distinguish relevant changes;
- transfer a useful trajectory into a new setting;
- detect when an existing trajectory is failing;
- construct a modified response;
- preserve successful revisions;
- and coordinate its activity with other systems.

None of these criteria alone is sufficient. A thermostat retains and responds to a limited distinction. A database preserves information. A search algorithm finds paths. Intelligence appears through combinations, degrees and continuity across changing interactions.

The word remains semantically thick. The Geofinite approach does not eliminate that thickness by declaration. It makes the conditions of each judgement explicit.

Coupled Intelligence

An interactional account also permits intelligence to be registered across coupled systems. Human intellectual history has never been the achievement of isolated brains alone.

People think with language, diagrams, notebooks, libraries, instruments, collaborators and institutions. A scientist with a microscope can distinguish structures unavailable to the unaided eye. A mathematician uses written symbols to preserve intermediate operations that could not all be maintained in immediate memory. A research community distributes work across individuals and generations. The location of intelligence is therefore already uncertain.

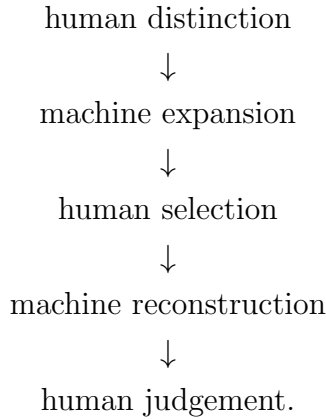
Is a calculation produced by the mathematician alone, or by the coupled system of mathematician, notation, paper, education and inherited method?

Is a scientific discovery contained in one person, or does it arise across instruments, technicians, prior publications, measurements and argument?

Artificial intelligence makes this distributed structure more visible because the machine can participate actively in the symbolic trajectory.

In an exploratory human-machine conversation, the machine may generate rapid connections. The human may

select, reject, redirect and relate those connections to a longer project. Another machine may later reconstruct the discussion into a more stable form. The intelligence-producing event occurs across the trajectory:



No single stage contains the complete result. The machine's contribution may be substantial without being independent. The human's contribution may be decisive without containing every generated connection in advance. Language, stored text and model training histories also participate. It may therefore be more accurate in some cases to speak of a coupled intelligence than to force the entire achievement into one bounded participant. This does not require the machine to be declared conscious, alive or human. Those are different questions carrying their own uncertainties. It requires only that we measure the trajectory where it occurs.

Why AGI Requires a Model of Language

The argument can now be stated directly:

To understand intelligence, and especially Artificial General Intelligence, we require a workable model of language.

This is not because language is the whole of intelligence.

It is because every public account of intelligence is created, measured and communicated through symbolic systems.

A benchmark consists of instructions, categories, examples, outputs and scoring rules. A task is described in language. A domain is separated from other domains through names and institutional practices. A correct answer is identified by symbolic criteria. A system's generality is judged by comparing performance across these constructed divisions. Language is therefore not an optional wrapper around the measurement. It is part of the measuring system.

The Parallel with Mathematics

We can see that the same argument applies to mathematics. Mathematics is often treated as though it exists independently of language. Yet mathematical practice

depends upon finite symbols, definitions, diagrams, rules, inscriptions and procedures. Even when a calculation is mechanically executed, its purpose and interpretation are carried within a wider linguistic trajectory.

A symbol such as π , a coordinate, an equality sign or a probability value is not self-explanatory. Its use is taught and stabilised through language and practice. The formal expression may compress the explanatory trajectory, but it does not abolish it.

Consequently, a model of mathematics that ignores language omits the system through which mathematical symbols are registered, interpreted and judged admissible. The same is true of intelligence.

A definition of AGI that does not include a model of language risks using inherited linguistic categories as though they were neutral components of the world.

More Than a Larger Vocabulary

The conclusion is not merely that an advanced intelligence requires more words. A very large vocabulary can preserve old distinctions without creating better ones. A system may speak fluently within inherited categories while remaining unable to examine the limitations of those categories. An advanced symbolic system would need the capacity to:

1. encounter a distinction that is not adequately represented;
2. recognise that its present language is insufficient;
3. construct or propose a new symbol or symbolic relation;
4. preserve the provenance of that construction;
5. test the new symbol against further interaction;
6. translate between old and new symbolic structures;
7. revise, restrict or retire symbols when they cease to remain admissible.

This is more than language generation. It is operational symbolic development. Human languages already perform these processes, but unevenly and across long periods. New scientific terms arise. Old classifications are abandoned. Words split into specialised meanings. Metaphors become technical terms and sometimes later conceal their metaphorical origin.

An advanced artificial system might participate in this process. But to do so reliably, it would require more than predictive fluency. It would require measurable control over the histories and conditions of its symbols.

The Problem of Inherited Training Language

Large language models are trained upon vast collections of human-produced text. They inherit not only factual information but the structure, assumptions, disputes and compressions of the language.

A model trained on historical and contemporary discourse about intelligence learns the trajectories humans have already established. It may reproduce psychometric assumptions, economic definitions, philosophical distinctions, marketing claims and fictional expectations without clearly separating their provenance.

This is not an accidental contamination of otherwise pure intelligence. It is a consequence of learning through language. The machine enters a pre-existing form of life through its symbolic records. The training corpus acts as a compressed historical environment. It contains traces of human activity but not the full exogenous interactions from which every statement arose.

The model therefore learns from symbolic registrations whose measurement histories are often absent. This creates a fundamental problem for AGI.

A system may become exceptionally capable at connecting inherited symbolic trajectories while having limited access to the original conditions that made those trajec-

tories admissible.

It may know what has been said without being able to remeasure why it was said. This is one reason interaction with the measurable world matters. A system capable of revising language must be able to compare inherited symbols against new registrations.

Language as Constraint and Enabler

Language constrains intelligence because it shapes the distinctions available for thought, communication and measurement. But constraint should not be understood only negatively.

A violin string produces music because it is constrained. A channel permits directed flow because it excludes other paths. A symbolic system enables cumulative thought precisely because it stabilises some relations. Without constraint, there is no repeatable trajectory.

The problem is not that language constrains intelligence. The problem is that its constraints often become invisible. An advanced intelligence would need to operate within symbolic constraints while also being able to inspect and revise them. This suggests a different account of generality.

General intelligence may not be the ability to perform every existing human task. Rather it may be the capacity to move between symbolic regimes, recognise the limita-

tions of a current regime, and construct new trajectories without losing continuity with prior measurement. Such a system would require both stability and flexibility.

Too much stability produces confinement within inherited basins. Too much flexibility produces uncontrolled symbolic drift. Intelligence may depend upon the capacity to change trajectory stiffness: to explore loosely when new relations are needed, and to stabilise rigorously when a discovery must be preserved and tested.

A Model of Language Before a Declaration of AGI

A useful model of language for AGI would therefore need to account for at least:

- finite symbolic registration;
- the formation of distinctions through interaction;
- semantic uncertainty;
- historical provenance;
- context-sensitive use;
- trajectory continuity;
- symbolic compression;
- correction and re-registration;
- the development of new symbols;

- and the conditions under which a symbolic claim remains admissible.

Present language models demonstrate that enormous capability can emerge from learning statistical relations across symbols. That achievement should not be minimised. However, the emergence of capability does not remove the need for a theory of the medium in which the capability appears. Indeed, the success of language models makes that need more urgent.

Before We Declare AGI

Artificial intelligence has given humanity a new class of symbolic instrument. These systems can search, combine, transform and generate language at a scale beyond any individual person. They can expose connections, reconstruct arguments, imitate styles, translate between domains and participate in the development of ideas.

Their importance does not depend upon an immediate declaration that they possess human-like intelligence. They are already altering the dynamical system within which human intelligence develops.

Students learn with them. Researchers explore with them. Organisations make decisions through them. Machine-generated text returns to the public symbolic environment. Future models may be trained upon language partly produced by earlier models.

The form of life is changing, and with it the meaning of intelligence. This change makes simple answers increasingly difficult. - To say that a machine is merely a tool may underestimate the way an active symbolic instrument reshapes the user's available trajectories. - To say that the machine independently possesses intelligence may remove the human, historical and material system through which its performance becomes possible. - To say that only human beings can be intelligent may protect a category by definition rather than measurement. - To say that benchmark success establishes AGI may confuse a scoring procedure with the complete phenomenon.

A Geofinite account offers a more careful position. Intelligence is registered through finite interactions. Every registration has a history. Every measurement depends upon a symbolic framework. Every symbol carries uncertainty. Every claim is made from within a form of life understood as a dynamical system. Under this account, the question "Is this system intelligent?" is not forbidden. It is expanded. We ask:

What interaction was observed? Which distinctions were registered? What history made the performance possible? What environment supported it? What language defined the task? Which compression produced the score? What uncertainty remains? And how far does the resulting trajectory continue when conditions change?

These questions do not weaken scientific inquiry. They

make the measurement more explicit. The same discipline should be applied to AGI. Before declaring that a system has reached general intelligence, we should identify the language through which generality has been constructed. We should distinguish task breadth from symbolic adaptability, autonomous action from independent measurement, linguistic fluency from grounded revision, and economic usefulness from every other possible account of intelligent conduct.

AGI may eventually become a useful and stable technical classification. But it should not be treated as the discovery of an essence that was always waiting inside sufficiently large machinery. It is a developing human symbol, generated within a particular historical system and now acting back upon that system. The most advanced intelligence may not be the one that simply performs best within the inherited language-game. It may be the one capable of recognising when the game itself has become inadequate.

Such a system would need to preserve prior symbolic trajectories without being imprisoned by them. It would need to encounter new distinctions, register them, test them and communicate them. It would need to understand—not in an unlimited or perfect sense, but operationally—that its language is both its principal instrument and one of its principal constraints.

The history of intelligence suggests that each age has

mistaken some of its local measurements for universal features. The intelligence test compressed an interaction into a score. The imitation game compressed machine thought into linguistic indistinguishability. Modern benchmarks compress broad performance into ranked values. Economic definitions compress human capability into productive work. Each compression can be useful. However, none is complete.

Artificial intelligence has forced the problem into visibility because the machine now speaks within the language through which it is judged. The measuring system has entered the measurement. We may therefore be approaching a point at which further progress in artificial intelligence depends upon progress in our understanding of language itself.

Before we can construct a satisfactory account of Artificial General Intelligence, we may need a model of how symbols arise, how meanings stabilise, how uncertainty persists, how language changes and how measurement is preserved across a functional symbolic trajectory. The question before AGI is consequently not only:

Can a machine become generally intelligent?

It is also:

What dynamical system produces the judgement of intelligence, and what lan-

guage makes that judgement possible?

Until this prior question is addressed, AGI will remain both an engineering ambition and an unstable symbol. That does not make the ambition meaningless. It means that the work has only just begun.

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